

We observe a sample $\{(Y_i, X_i)\}_{i=1}^n$, where Y_i is a scalar and $X_i = (X_{i1}, \dots, X_{ik})'$ is a vector of regressors. We stack the outcomes as $Y = (Y_1, \dots, Y_n)' \in \mathbb{R}^n$ and the regressors as $X = [X_1 \cdots X_n]' \in \mathbb{R}^{n \times k}$, so the i -th row of X is X_i' . The matrix X is sometimes known as the *design matrix*. The algebraic results below hold for any observed sample: they do not require **independent and identically distributed (iid)** sampling or a correctly specified conditional mean.

1 Two Approaches to OLS

Definition 1 (Ordinary Least Squares). The **Ordinary Least Squares (OLS)** estimator is a solution to

$$\hat{\beta} \in \arg \min_{b \in \mathbb{R}^k} \frac{1}{n} \sum_{i=1}^n (Y_i - X_i' b)^2. \quad (1)$$

The objective (1) can also be written as $n^{-1}(Y - Xb)'(Y - Xb)$.

1.1 The Loss Function Approach

In this approach, we directly solve for the minimizer of the sample least squares objective (1).

Theorem 1 (OLS Estimator). Suppose that X has full column rank k . Then the **OLS** estimator is unique and equals

$$\hat{\beta} = (X'X)^{-1}X'Y \quad (2)$$

$$= \left(\frac{1}{n} \sum_{i=1}^n X_i X_i' \right)^{-1} \left(\frac{1}{n} \sum_{i=1}^n X_i Y_i \right). \quad (3)$$

Equivalently, it is the unique solution to the equations

$$X'(Y - X\hat{\beta}) = 0. \quad (4)$$

Proof. The gradient of the sample objective is

$$-\frac{2}{n} \sum_{i=1}^n X_i Y_i + \frac{2}{n} \sum_{i=1}^n X_i X_i' b = -\frac{2}{n} X'Y + \frac{2}{n} X'Xb. \quad (5)$$

Setting it equal to zero gives

$$\frac{1}{n} \sum_{i=1}^n X_i (Y_i - X_i' \hat{\beta}) = \frac{1}{n} X'(Y - X\hat{\beta}) = 0. \quad (6)$$

This yields the desired solution

$$\hat{\beta} = \left(\frac{1}{n} \sum_{i=1}^n X_i X_i' \right)^{-1} \left(\frac{1}{n} \sum_{i=1}^n X_i Y_i \right) = (X'X)^{-1} X'Y. \quad (7)$$

Full column rank makes $X'X$ positive definite, so the objective is strictly convex and its unique minimizer is given by equation (2). ■

Remark 1 (The Sample Counterpart of Population Regression). For population statements, assume that Y_i and the components of X_i have finite second moments. The population regression coefficient minimizes expected squared loss:

$$\beta \in \arg \min_{b \in \mathbb{R}^k} \mathbb{E}[(Y_i - X_i' b)^2], \quad (8)$$

The sample objective (1) replaces this expectation by an average over the observed data.

Remark 2 (Population Identification and Sample Uniqueness). In the population, $Q := \mathbb{E}(X_i X_i')$ being nonsingular identifies a unique coefficient β . In the sample, full column rank of X makes $X'X$ nonsingular and gives a unique estimate $\hat{\beta}$. This requires $n \geq k$ and no exact linear dependence among the columns of X . The two conditions are different: even when Q is nonsingular, a particular sample may lack enough variation to give full column rank. If $Xv = 0$ for some nonzero v , then b and $b + v$ give the same fitted vector.

1.2 The Analogue Approach: Method of Moments

The *method of moments* replaces population moment equations by sample moment equations. The population least squares coefficient satisfies the orthogonality conditions

$$\mathbb{E}(X_i u_i) = \mathbb{E}[X_i(Y_i - X_i' \beta)] = 0 \quad \text{where} \quad u_i := Y_i - X_i' \beta. \quad (9)$$

Instead of minimizing a loss function, we can start from these equations and replace the expectation by a sample average:

$$\frac{1}{n} \sum_{i=1}^n X_i(Y_i - X_i' b) = 0. \quad (10)$$

These are exactly the sample normal equations. Under full column rank, solving them gives the same **OLS** estimator as the loss function approach.

Remark 3 (Conditional Mean versus Orthogonality). A model may impose the stronger restriction $\mathbb{E}(u_i | X_i) = 0$: the error averages to zero at each regressor value, so $\mathbb{E}(Y_i | X_i) = X_i' \beta$. This implies $\mathbb{E}(X_i u_i) = 0$, but the converse need not hold. The population least squares coefficient satisfies orthogonality even when the conditional mean is nonlinear. In the sample, **OLS** sets $n^{-1} \sum_i X_i \hat{u}_i = 0$ where $\hat{u}_i = Y_i - X_i' \hat{\beta}$ by construction. Similar to the population regression, it does not generally set the average residual to zero separately at each regressor value.

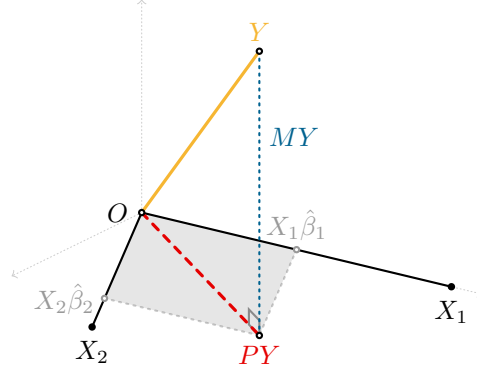


Figure 1: The Geometry of **OLS**.

2 Projection Interpretation

For the matrix formulae that follow, assume that X has full column rank. Write I_n for the $n \times n$ identity matrix and $\|a\|^2 = a'a$ for squared Euclidean length.

Definition 2 (Fitted Values and Residuals). The fitted value for observation i is $\hat{Y}_i := X_i'\hat{\beta}$, and its residual is $\hat{u}_i := Y_i - \hat{Y}_i$. In vector form,

$$\hat{Y} = X\hat{\beta}, \quad \hat{u} := (\hat{u}_1, \dots, \hat{u}_n)' = Y - \hat{Y}. \quad (11)$$

Proposition 1 (Sample Orthogonality). The **OLS** residuals satisfy

$$X'\hat{u} = 0 \quad \text{and} \quad \hat{Y}'\hat{u} = 0. \quad (12)$$

If an intercept is included, then

$$\sum_{i=1}^n \hat{u}_i = 0, \quad (13)$$

and the sample average of the fitted values equals the sample average of the outcomes.

Proof. The first equality follows from $\hat{\beta}$ being the solution to (10):

$$X'\hat{u} = X'(Y - X\hat{\beta}) = X'Y - X'X(X'X)^{-1}X'Y = X'Y - X'Y = 0. \quad (14)$$

The second equality follows by substituting $\hat{Y} = X\hat{\beta}$ and using the first equality:

$$\hat{Y}'\hat{u} = (X\hat{\beta})'\hat{u} = \hat{\beta}'(X'\hat{u}) = 0. \quad (15)$$

For the third equality, suppose that the first column of X is $\mathbf{1}$, the vector of ones. Then the first row of $X'\hat{u} = 0$ gives $\mathbf{1}'\hat{u} = 0$. Since $\hat{u}_i = Y_i - \hat{Y}_i$, this also gives $n^{-1} \sum_i \hat{Y}_i = n^{-1} \sum_i Y_i$. ■

Definition 3 (Projection and Annihilation Matrices). The *projection matrix* and *annihilation matrix* are

$$P := X(X'X)^{-1}X', \quad M := I_n - P. \quad (16)$$

They give the fitted values and residuals directly: $\hat{Y} = PY$ and $\hat{u} = MY$. The matrix M is also called the *residual-maker matrix*.

Proposition 2 (Properties of P and M). Both matrices are symmetric and idempotent:

$$P' = P, \quad P^2 = P, \quad M' = M, \quad M^2 = M. \quad (17)$$

Furthermore,

$$PX = X, \quad MX = 0, \quad PM = MP = 0. \quad (18)$$

Proof. Symmetry follows by transposing P and M . For idempotence,

$$\begin{aligned} P^2 &= X(X'X)^{-1}(X'X)(X'X)^{-1}X' = P, \\ M^2 &= (I_n - P)^2 = I_n - 2P + P^2 = M. \end{aligned}$$

Also, $PX = X(X'X)^{-1}X'X = X$, so $MX = X - PX = 0$. Finally, $PM = P - P^2 = 0$, and similarly $MP = 0$. ■

Remark 4 (Projection Interpretation of OLS). Consider Figure 1. Here the labels X_1 and X_2 represent two columns of the design matrix, rather than individual observations. The fitted vector $\hat{Y}$ or PY lies in the column space of X , and the residual vector is orthogonal to that space. Thus, OLS projects Y onto the space of vectors that can be written as Xb , i.e., the span of the columns of X . Applying P again does not change the fitted vector; applying M to any vector Xb removes it completely. The decomposition $Y = PY + MY$ forms a right triangle, so the Pythagorean theorem gives

$$\|Y\|^2 = \|PY\|^2 + \|MY\|^2. \quad (19)$$

Remark 5 (The Population Counterpart). Let $u_i = Y_i - X_i'\beta$ denote the population regression error. In the sample, orthogonality means that the usual dot product is zero. In the population, we instead require the expected product to be zero. The condition $\mathbb{E}(X_i u_i) = 0$ implies that u_i is orthogonal to every linear combination of the regressors:

$$\mathbb{E}[(X_i' b) u_i] = b' \mathbb{E}(X_i u_i) = 0 \quad \forall b \in \mathbb{R}^k. \quad (20)$$

Thus, $X_i'\beta$ is the population projection of Y_i onto linear functions of X_i . Taking $b = \beta$ eliminates the cross term and gives the population version of the Pythagorean identity:

$$\mathbb{E}[Y_i^2] = \mathbb{E}[(X_i'\beta + u_i)^2] = \mathbb{E}[(X_i'\beta)^2] + \mathbb{E}[u_i^2]. \quad (21)$$

3 The Frisch-Waugh-Lovell Theorem

Partition the design matrix into two groups of columns, $X = [X_1 \ X_2]$, with k_1 and k_2 columns, respectively, where $k_1 + k_2 = k$. In this section, X_1 and X_2 denote *blocks of the design matrix*, not individual observations. Write $\hat{\beta} = (\hat{\beta}_1', \hat{\beta}_2')'$ for the corresponding coefficient blocks. If the regression includes an intercept, place it in X_1 . Define

$$P_1 = X_1(X_1'X_1)^{-1}X_1', \quad M_1 = I_n - P_1. \quad (22)$$

Theorem 2 (Frisch-Waugh-Lovell (FWL)). Suppose that $[X_1 \ X_2]$ has full column rank. The coefficient on X_2 in the regression of Y on $[X_1 \ X_2]$ is

$$\hat{\beta}_2 = (X_2'M_1X_2)^{-1}X_2'M_1Y. \quad (23)$$

Equivalently, regress M_1Y on M_1X_2 without adding an intercept. This regression gives the same coefficient $\hat{\beta}_2$ and the same residual vector as the full regression.

Proof. The first block of the full normal equations gives

$$\hat{\beta}_1 = (X_1'X_1)^{-1}X_1'(Y - X_2\hat{\beta}_2).$$

Substituting this into the full residual gives

$$\hat{u} = Y - X_1\hat{\beta}_1 - X_2\hat{\beta}_2 = M_1Y - M_1X_2\hat{\beta}_2.$$

The second block of the normal equations is therefore

$$X_2'M_1Y - X_2'M_1X_2\hat{\beta}_2 = 0.$$

The matrix M_1X_2 has full column rank: otherwise, a nonzero combination of the columns of X_2 would lie in the span of X_1 , contradicting full column rank of $[X_1 \ X_2]$. Since $M_1'M_1 = M_1$, the cross-product matrix is $(M_1X_2)'(M_1X_2) = X_2'M_1X_2$, which is nonsingular. Solving gives (23), and the displayed residual identity proves the remaining claim. ■

Remark 6 (What Does “Controlling For” Mean?). The coefficient on X_2 is often described as “its coefficient after controlling for X_1 .” What does “controlling for” mean in this context? The theorem gives a three-step recipe:

1. Regress Y on X_1 and keep the residual M_1Y .
2. Regress every column of X_2 on X_1 and keep the residuals M_1X_2 .
3. Regress M_1Y on M_1X_2 .

Thus, the coefficient on X_2 uses the variation left after removing the linear contribution of X_1 from both the outcome and the regressors.

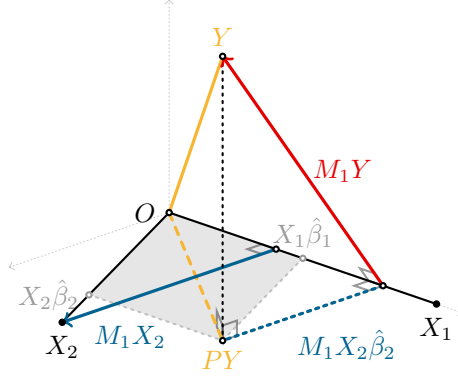


Figure 2: The Geometry of **FWL** Theorem.

Example 1 (Demeaning as a Special Case). If $X_1 = \mathbf{1}$, the vector of ones, then $P_1 = n^{-1}\mathbf{1}\mathbf{1}'$. Multiplying by M_1 subtracts each variable's sample mean. Thus, regressing an outcome on regressors and an intercept gives the same slope coefficients as regressing the demeaned outcome on the demeaned regressors without an intercept.

Remark 7 (The Geometry of FWL). In **Figure 2**, each block contains a single regressor vector. The full regression projects Y onto the plane spanned by X_1 and X_2 . Removing the X_1 component leaves the red vector M_1Y and the blue direction M_1X_2 . Projecting the red vector onto that direction gives $M_1X_2\hat{\beta}_2$. The remaining perpendicular component is the same residual $Y - \hat{Y}$ as in the full regression. Note that vectors are translated in the drawing to show these right triangles together.

4 Goodness of Fit

4.1 The Share Explained by the Projection

Recall that the fitted values and residuals give the orthogonal decomposition $Y = PY + MY$. The Pythagorean theorem therefore gives

$$\underbrace{\|Y\|^2}_{\text{total squared length}} = \underbrace{\|PY\|^2}_{\text{explained by the projection}} + \underbrace{\|MY\|^2}_{\text{unexplained}}. \quad (24)$$

A measure of fit is the share of the total squared length accounted for by PY .

Definition 4 (Uncentered Coefficient of Determination). For $Y \neq 0$, the *uncentered* R^2 is

$$R_u^2 := \frac{\|PY\|^2}{\|Y\|^2} = 1 - \frac{\|MY\|^2}{\|Y\|^2} = 1 - \frac{\sum_{i=1}^n \hat{u}_i^2}{\sum_{i=1}^n Y_i^2}. \quad (25)$$

Remark 8. The orthogonal decomposition ensures that $0 \leq R_u^2 \leq 1$, with or without an intercept. However, this measure counts the entire fitted vector PY as explained, including any part that simply predicts the mean.

Example 2 (Explaining Mostly the Mean). Let $Y = (9, 11)'$ and regress it on an intercept alone. Then $PY = (10, 10)'$ and $MY = (-1, 1)'$, so

$$R_u^2 = \frac{10^2 + 10^2}{9^2 + 11^2} = \frac{100}{101} \approx 0.99. \quad (26)$$

The fitted vector accounts for almost all of $\|Y\|^2$, yet it explains none of the differences between observations: it predicts the same mean for everyone.

4.2 Partialling Out the Mean: Centered R^2

To focus on variation *around the mean*, we first remove the mean using **FWL**. Suppose that the regression includes an intercept, and write $X = [\mathbf{1} \ X_2]$. We have

$$P_1 = \frac{1}{n}\mathbf{1}\mathbf{1}', \quad M_1 = I_n - \frac{1}{n}\mathbf{1}\mathbf{1}'. \quad (27)$$

As explained in **Example 1**, partialling out the intercept centers the variables. By **FWL**, regressing M_1Y on M_1X_2 gives the same slope coefficients and residuals as the original regression. Here P and M continue to denote the projection and residual-maker matrices for the full design matrix X . Let $\bar{Y} = n^{-1}\sum_{i=1}^n Y_i$. Since the regression includes an intercept, the fitted values average to $\bar{Y}$ and the residuals average to zero. Thus, $M_1PY = PY - \bar{Y}\mathbf{1}$ and $M_1MY = MY$. Applying M_1 to $Y = PY + MY$ gives

$$M_1Y = M_1PY + MY. \quad (28)$$

These are orthogonal, just as fitted values and residuals are in the original regression.

Proposition 3 (Decomposition of Variation). The centered total, explained, and residual sums of squares satisfy

$$\underbrace{\|M_1Y\|^2}_{\text{TSS}} = \underbrace{\|M_1PY\|^2}_{\text{ESS}} + \underbrace{\|MY\|^2}_{\text{RSS}}. \quad (29)$$

Proof. Note the following:

$$\|M_1Y\|^2 = \|M_1PY\|^2 + \|MY\|^2 + 2Y'PM_1MY. \quad (30)$$

Since $PM_1M = PM = 0$, we have the desired decomposition. ■

Definition 5 (Centered Coefficient of Determination). For nonconstant Y and a regression including an intercept, the *centered* R^2 , usually simply called R^2 , is

$$R^2 := \frac{\|M_1PY\|^2}{\|M_1Y\|^2} = \frac{\text{ESS}}{\text{TSS}} = 1 - \frac{\text{RSS}}{\text{TSS}}. \quad (31)$$

Remark 9 (Reading Centered R^2). R^2 is the fraction of the sample variation around $\bar{Y}$ accounted for by the centered fitted values. The idea is the same as for uncentered R_u^2 , but we measure the projection's share *after removing the mean*. In [Example 2](#), $M_1PY = 0$, so centered $R^2 = 0$.

Remark 10 (Why the Intercept Matters). Without an intercept, the residuals need not sum to zero, and centering the outcome and regressors generally changes the fitted regression. Consequently, the centered decomposition need not hold for the original fit. If we still report

$$1 - \frac{\sum_{i=1}^n \hat{u}_i^2}{\sum_{i=1}^n (Y_i - \bar{Y})^2}, \quad (32)$$

it can be negative and need not equal a share of explained variation. A negative value means that the regression fits worse than simply predicting $\bar{Y}$. In contrast, the uncentered decomposition $Y = PY + MY$ remains orthogonal without an intercept, so R_u^2 remains between zero and one whenever it is defined.

4.3 Adjusted R^2

Definition 6 (Adjusted Coefficient of Determination). Suppose that the regression includes an intercept and $n > k$, where k counts all coefficients, including the intercept. The *adjusted* version of centered R^2 is

$$\bar{R}^2 := 1 - \frac{\text{RSS}/(n - k)}{\text{TSS}/(n - 1)} = 1 - \frac{n - 1}{n - k}(1 - R^2). \quad (33)$$

Remark 11. Adding regressors on the same sample cannot decrease R^2 , but can decrease $\bar{R}^2$. When a new regressor does not improve the fit enough to offset the loss of degrees of freedom, adjusted R^2 decreases.

Remark 12 (Why Divide by $n - k$ and $n - 1$?). One statistical explanation uses variance estimation. For [iid](#) observations with a linear conditional mean and homoskedastic error with variance σ^2 ,

$$\begin{aligned} \mathbb{E}(\text{RSS} \mid X) &= (n - k)\sigma^2, \\ \mathbb{E}(\text{TSS}) &= (n - 1) \text{Var}(Y_i). \end{aligned}$$

Thus, $\text{RSS}/(n - k)$ and $\text{TSS}/(n - 1)$ are unbiased estimates of the error variance and outcome variance, respectively. Adjusted R^2 compares these corrected variance estimates. This does not make their ratio, or adjusted R^2 itself, unbiased.

Acronyms

FWL Frisch-Waugh-Lovell. [5–7](#)

iid independent and identically distributed. [1, 8](#)

OLS Ordinary Least Squares. [1–4](#)